

C: > Users > Afroz > Desktop > EXP4 > exp4.c

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  char input[100];
6  int pos = 0;
7
8  void E();
9  void Eprime();
10 void T();
11 void Tprime();
12 void F();
13
14 void error()
15 {
16     printf("Invalid Expression\n");
17     exit(0);
18 }
19
20 void E()
21 {
22     T();
23     Eprime();
24 }
25
26 void Eprime()
27 {
28     if (input[pos] == '+')
29     {
30         pos++;
31         T();
32         Eprime();
33     }
34     else if (input[pos] == '-')
35     {
36         pos++;
37         T();
38         Eprime();
39     }
40 }
41
42 void T()
43 {
44     F();
45     Tprime();
46 }
47
48 void Tprime()
49 {
50     if (input[pos] == '*')
51     {
52         pos++;
53         F();
54         Tprime();
55     }
56     else if (input[pos] == '/')
57     {
58         pos++;
59         F();
60         Tprime();
61     }
62 }
63
64 void F()
65 {
66     if (input[pos] >= '0' && input[pos] <= '9')
67     {
68         while (input[pos] >= '0' && input[pos] <= '9')
69             pos++;
70     }
71     else if (input[pos] == '(')
72     {
73         pos++;
74         E();
75
76         if (input[pos] == ')')
77             pos++;
78         else
79             error();
80     }
81     else
82     {
83         error();
84     }
85 }
86
87 int main()
88 {
89     printf("Enter arithmetic expression: ");
90     scanf("%s", input);
91
92     E();
93
94     if (input[pos] == '\0')
95         printf("Valid Expression\n");
96     else
97         printf("Invalid Expression\n");
98
99     return 0;
100 }
101
```

```
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP4
$ nano exp4.c

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP4
$ gcc exp4.c -o exp4

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP4
$ ./exp4
Enter arithmetic expression: 2+3*4
Valid Expression

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP4
$ ./exp4
Enter arithmetic expression: 2+*3
Invalid Expression

Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP4
$ |
```